SEMESTER-II

UBT009: BIOCHEMISTRY-I

L	T	P	Cr
3	0	0	3.0

Course Objective: The students will know how the collection of thousands inanimate molecules that constitute living organisms interact to maintain and perpetuate life governed solely by the physical and chemical laws as applicable to the nonliving thing.

Syllabus

Biochemistry: Introduction as a discipline-historical perspective, major landmarks in the development of biochemistry.

Chemical Foundations of living systems: Molecular basis of life, Biological chemistry – Biomolecules, Metabolism – Basic concepts and Design, Bioenergetics- Entropy, Biochemical equilibria, Dissociation and association constants, pH and buffers.

Interactions in biological systems: Intra and intermolecular forces, Electrostatic and hydrogen bonds, Disulfide bridges, Hydrophobic and hydrophilic molecules and forces, Water and weak interactions, Debye-Huckel Theory.

Carbohydrates: Classification, Monosaccharides – structures and function; reactions of monosaccharides- mutarotation, glycoside formation, reduction and oxidation, epimerization and esterification, polarimetry; important monosaccharides and disaccharide; Polysaccharides –overview, structure; important polysaccharide; plant polysaccharide; Glycosaminoglycans, Glycoproteins.

Amino acids and Proteins: Amino acids as building blocks of proteins, their structure, classification and chemical properties; non- proteinogenicaminoacids; Structure of peptide bond, organizational levels of protein structure; alpha- helix, beta pleated sheet, Ramachandran Plot.

Nucleic Acids and Porphyrins: Structure and properties of nucleic acid bases, nucleosides and nucleotides, biologically important nucleotides, Physical and chemical properties of RNA/DNA. Hydrolysis of nucleic acids. Structure, properties and classification of porphyrins.

Lipids: Fatty acids as building blocks of most lipids, their structure and properties, classification of lipids, General structure and function of major lipid subclasses: Acylglycerols, phosphoglycerides, sphingolipids, glycosphingolipids, terpenes, steroids, Prostaglandins.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Know the chemical constituents of cells, the basic units of living organisms.
2. Explain various types of weak interactions between the biomolecules.
3. Know how the simple precursors give rise to large biomolecules such as proteins, carbohydrates, lipids, nucleic acids.
4. Correlate the structure-function relationship in various biomolecules
5. Know the role of biomolecules for orderly structures of the cells/tissues.

Text Books

1. Nelson, DL and Cox MM., Lehninger: Principles of Biochemistry, WH Freeman (2008) 5th ed.
2. David E Metzler: Biochemistry, The Chemical reactions of Living Cells Vol. 1. 2nd Edition, Elsevier Academic Press (2003),
3. Berg JM, Tymoczko JL and Stryer L: Biochemistry, 5th Edition, WH Freeman and Company, (2005)

Reference Books

1. Koolman J and Roehm K H Color: Atlas of Biochemistry, 2nd Edition, Georg Thieme Verlag Publishers (2005)
2. Jain, JL, Jain, S and Jain, N: Fundamentals of Biochemistry, S. Chand and Company Ltd. (2005).
3. Plummer DT: An Introduction to Practical Biochemistry, Tata McGraw-Hill Publishing Company Limited (1988)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30